wide array of complex reasoning benchmarks). This comparison helps determine whether advanced, broad reasoning skills on well-trained models even with exceptionally large-scale can inherently address complex temporal tasks.

4.3 Experimental Setup

Implementation. All our experiments build upon Qwen2.5-3B-Instruct [42], a moderate size for fast adaptability and cost efficiency. We implement our three-stage RL fine-tuning framework using veRL framework [45], adopting the GRPO algorithm detailed in Equation (3). All RL fine-tuning experiments were conducted on four NVIDIA A6000 GPUs.

Hyperparameters. Key hyperparameters for the GRPO optimization include KL coefficient $\beta = 0.001$, and $K = 5$ rollout responses per prompt for group-normalized advantage estimation. The full configuration details can be found at Appendix A.

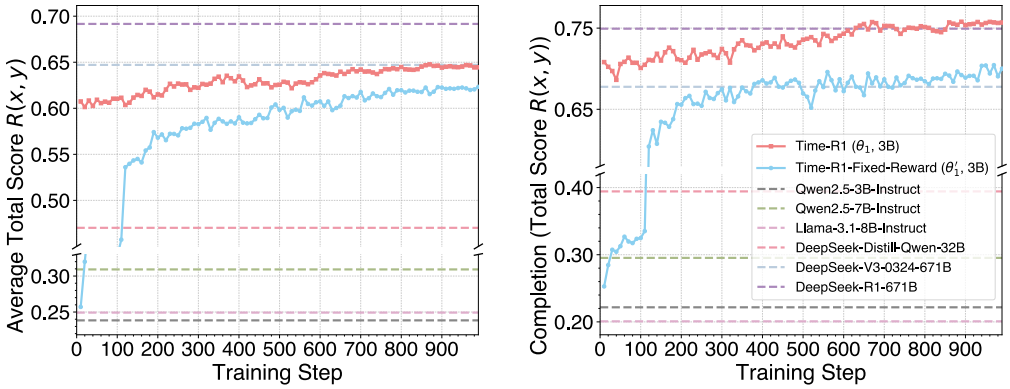
4.4 Main Results

We now present the core experimental results, evaluating the performance of Time-R1 across its training stages against the established baselines. We specifically report on the performance of the model checkpoint after Stage 1 (θ_1) for foundational tasks and the checkpoint after Stage 2 (θ_2) for future prediction and scenario generation.

4.4.1 Stage 1: Foundational Temporal Reasoning Performance

The effectiveness of our Stage 1 fine-tuning on core temporal understanding is demonstrated by the training dynamics in Figure 3 (see appendix C for details of fine-tuning curves for all subtasks and phases) and the final scores in Table 1. The results highlight the substantial benefits of our Stage 1 RL fine-tuning. Time-R1 (θ_1) demonstrates a remarkable improvement in its overall average score, with an increase of approximately 171.6% over its base Qwen2.5-3B-Instruct model.

Significantly, with these improvements, Time-R1 now outperforms the much larger DeepSeek-V3-0324-671B model and is highly competitive with the state-of-the-art 671B DeepSeek-R1 model. It secures the top performance on the demanding Completion task and the second-best performance on the challenging Event Ordering task. This strong performance, rivaling or exceeding much larger baselines, is largely attributed to our meticulously designed task-specific rewards and the dynamic reward curriculum. For instance, the inconsistency and diversity penalties for Event Ordering (detailed in Section 3.3.2) are pivotal. The learning curves in Appendix C also illustrate that the



(a) Overall Total Score $R(x, y)$ (b) Completion (Total Score $R(x, y)$)

Figure 3: **Stage 1 Training Performance *vs.* Baselines.** Training curves for Time-R1 (θ_1) and its ablation variant, Time-R1-Fixed-Reward (θ_1'), evaluated against baseline models (indicated by horizontal dashed lines). Plot (a) shows the Overall Total Score across all subtasks, while plot (b) presents the Masked Time Entity Completion subtask. The solid lines demonstrate our models' scores improving throughout the training process, ultimately surpassing the performance levels of most baseline models, including those with significantly larger scales.

Table 1: **Stage 1 Foundational Temporal Reasoning Performance.** Average Total Score ($R(x, y)$) on the four subtasks and overall. Higher scores indicate better performance. Best score in each column is **bold**, second best is underlined.

Model	Overall Avg. ↑	Ordering ↑	Completion ↑	Inference ↑	Difference ↑
Qwen2.5-3B-Instruct	0.2384	0.1583	0.2217	0.3372	0.2363
Qwen2.5-7B-Instruct	0.3092	0.2775	0.2953	0.3366	0.3275
Llama-3.1-8B-Instruct	0.2492	0.2239	0.2008	0.3339	0.2383
DeepSeek-Distill-Qwen-32B	0.4702	0.5026	0.3943	0.5264	0.4576
DeepSeek-V3-0324-671B	0.6471	0.6409	0.6777	0.6796	0.5901
DeepSeek-R1-671B	**0.6916**	**0.6848**	0.7493	**0.7145**	**0.6172**
Time-R1-Fixed-Reward (θ_1', 3B)	0.6259	0.6623	0.6977	0.5813	0.5621
Time-R1 (θ_1, 3B)	0.6476	0.6815	**0.7555**	0.5938	0.5599

model's adherence to response consistency and diversity for this task steadily improves, reflecting enhanced logical reasoning. Such effective instillation of logical mapping allows Time-R1 to compete effectively with much larger models on these complex temporal logic challenges.

To validate the contribution of our reward design, we include an ablation model, **Time-R1-Fixed-Reward (θ_1')**, which was trained using a static, strict reward function. As shown in Figure 3, the full Time-R1 model consistently outperforms this ablation variant, underscoring the importance of the dynamic curriculum, which will be analyzed further in Section 4.5.1.

4.4.2 Stage 2: Future Event Time Prediction

Stage 2 equips models to predict event timing post-knowledge cutoff (2023). We assess our full pipeline and Stage 1's impact by evaluating two variants: **Time-R1 (θ_2, 3B)** (full curriculum, Section 3.2) and an ablation model, **Time-R1-S2-Direct (θ_2', 3B)** (Stage 2 fine-tuning only, from base Qwen2.5-3B-Instruct, omitting Stage 1). Performance is compared against baselines for August 2024 - February 2025 predictions.

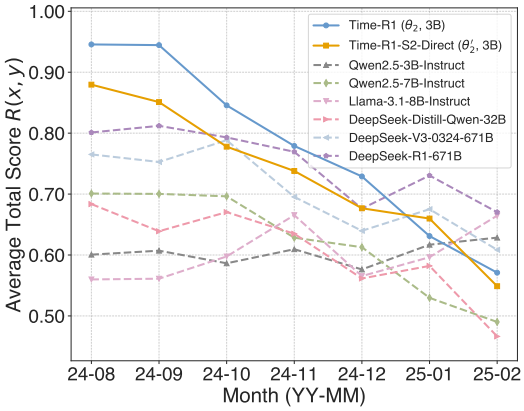


Figure 4: Monthly Average Total Score $R(x, y)$ for Stage 2 Future Event Prediction (August 2024 - Feb 2025). Compares Time-R1 variants (θ_2 and θ_2') against baselines. Evaluated with $\alpha = 0.1$.

The overall Stage 2 performance, measured by Average Total Score $R(x, y)$ with strict evaluation ($\alpha = 0.1$), is presented in Table 2. While models show clear improvement over Stage 1 Inference tasks, likely aided by a narrower prediction time span, further significant gains prove challenging. For instance, the DS-Qwen-32B model, despite its scale and specialized complex reasoning training, scores lower than some 3B models lacking such enhancements (*e.g.*, the base Qwen2.5-3B-Instruct), underscoring the inherent difficulty of learning extrapolation and handling post-cutoff data.Our primary model, Time-R1 (θ_2, 3B), achieves the highest score. This strong performance, consistent across the prediction horizon (Figure 4), shows it generally outperforming most baselines, including the much larger DeepSeek-R1-671B and DeepSeek-V3-671B models. This robust result strongly supports our hypothesis that specialized, staged temporal fine-tuning enables smaller models to achieve superior performance on challenging future prediction tasks. Furthermore, these findings highlight general LLM weaknesses in temporal reasoning and underscore the efficacy and necessity of our structured training framework. The foundational understanding from Stage 1, combined with Stage 2's predictive skill development, underpins this strong near-future temporal reasoning (see Section 5.2 for challenges facing standard LLMs). The ablation model, Time-R1-S2-Direct (θ_2', 3B), also demonstrates solid performance, outperforming several baselines and indicating Stage 2 RL fine-tuning's standalone effectiveness. See more discussion on Section 4.5.2.

Table 2: **Stage 2 Future Event Prediction Performance (Overall).** Average Total Score $R(x, y)$ evaluated with $\alpha = 0.1$. Higher scores are better. Best score is **bold**, second best is underlined. θ_2 checkpoint of Time-R1 is used.

Metric	Qwen2.5 -3B	Qwen2.5 -7B	Llama-3.1 -8B	DS-Qwen -32B	DS-V3 -671B	DS-R1 -671B	**Time-R1 (θ_2', 3B)**	**Time-R1 (θ_2, 3B)**
Avg. Total Score ↑	0.6036	0.6226	0.6015	0.5997	0.7036	0.7503	0.7331	**0.7780**

4.4.3 Stage 3: Creative Scenario Generation Quality

Finally, we evaluate model generalization to generating plausible future scenarios—a task without explicit fine-tuning. Table 3 presents AvgMaxSim scores, quantifying the semantic plausibility of generated news scenarios against real news events (August 2024 - February 2025). Results demonstrate Time-R1 (θ_2, 3B)'s strong generalization capability. It achieves the highest overall AvgMaxSim score, surpassing all baseline models, including the very large DeepSeek-V3-0324-671B and DeepSeek-R1-671B. Monthly scores for Time-R1 (θ_2, 3B) also reveal consistently strong performance. This Stage 3 success, achieved without direct training on generation, underscores the S1+S2 curriculum's effectiveness in building robust, transferable temporal reasoning. These capabilities are significant for addressing research gaps in challenging future prediction and generation tasks and demonstrate practical application value. Our ablation model, Time-R1-S2-Direct (θ_2', 3B), also performs commendably, outperforming some baselines (further discussion in Section 4.5.2).

Table 3: **Stage 3 Creative Scenario Generation Plausibility (AvgMaxSim Scores (%)).** Compares semantic similarity of generated scenarios to real news events (August 2024 - Feb 2025). Higher scores indicate better plausibility. Best overall average is **bold**, second best is underlined.

Model	**Avg. (%)**	**Monthly AvgMaxSim Scores (%)**						
	↑	24-08	24-09	24-10	24-11	24-12	25-01	25-02
Qwen2.5-3B-Instruct	47.66	47.27	46.89	47.39	48.57	48.77	47.76	46.94
Qwen2.5-7B-Instruct	47.59	46.99	49.78	46.18	48.53	46.91	48.88	45.83
Llama-3.1-8B-Instruct	47.96	48.99	50.03	47.42	46.21	47.06	48.01	48.03
DeepSeek-Distill-Qwen-32B	47.12	46.58	46.78	47.94	47.04	48.40	47.30	45.81
DeepSeek-V3-0324-671B	48.81	50.73	51.77	48.60	48.46	47.52	47.71	46.85
DeepSeek-R1-671B	47.46	47.55	49.64	47.29	45.29	47.85	47.30	47.31
Time-R1-S2-Direct (θ_2', 3B)	47.93	47.89	47.11	47.95	48.29	46.05	50.69	47.52
Time-R1 (θ_2, 3B)	**48.90**	47.75	48.29	49.81	48.77	49.03	50.81	47.83

4.5 Ablation Studies

4.5.1 Impact of Dynamic Reward Mechanism

Our methodology (Section 3.3.3) employs a dynamic reward mechanism during Stage 1 fine-tuning. This curriculum learning approach, with its phased adjustment of reward strictness (from lenient $\alpha_{\text{start}} = 0.07$ to strict $\alpha_{\text{target}} = 0.1$), is designed to mitigate cold-start challenges and guide the model towards robust performance on complex temporal tasks. We hypothesized this would lead to superior learning compared to a static, strict reward function.

The empirical results presented in Figure 3 validate this hypothesis. The advantage of the dynamic reward curriculum is evident both in the Overall Total Score across all subtasks (Figure 3a) and in the specific Masked Time Entity Completion subtask (Figure 3b). For the overall performance, the full Time-R1 model achieves a consistently higher and more stable score than the fixed-reward ablation model. This performance gap is even more pronounced in the Completion subtask, where the fixed-reward model's progress begins to slow and plateau around a score of 0.70. In contrast, the curriculum-trained model continues to improve, achieving a significantly higher and more stable final score of over 0.75. This suggests that the curriculum's initial leniency and gradual transition to stricter evaluation criteria enable more effective exploration and learning, preventing convergence to a sub-optimal policy and leading to a better mastery of the task.